

A CLASS Benchmark for Late-Time Matter-Power Suppression from a Fold-Triggered Dark-Matter Mass Drain

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Abstract

We present a CLASS implementation benchmark for a fold-triggered late-time dark-matter mass-drain mechanism, referred to below as Selection-Fold Dark Drain. In this mechanism a fold transition in a branch variable u entrains a scalar $x = \theta/M_{\text{Pl}}$ and changes the cold-dark-matter mass as $m_c(x) = m_{c0} \exp[-\beta_x(x - x_{\text{ini}})]$. The resulting late-time matter drain is tested through a sequence of background, source-level, response-level, and Newtonian-gauge perturbation checks.

The background bridge reproduces the reconstructed Friedmann branch with $\max |d \ln H / dN - d \ln E_{\text{Mo}} / dN| \simeq 7.68 \times 10^{-10}$. The final matter-source insertion point is verified by a control test in which a half-amplitude source shift gives $P_{\text{on}}/P_{\text{off}} = 0.25$. In the fiducial Newtonian-gauge branch we find $P_{\text{geo}} \simeq 0.8971$, $P_{\text{min}} \simeq 0.8896$, $P_{\text{max}} \simeq 0.9116$, and a residual shape variation tilt $\simeq 1.61\%$ over the tested modes.

Interpreted as an amplitude proxy, this corresponds to $\sqrt{P_{\text{geo}}} \simeq 0.947$ and hence to a roughly five-percent S_8 -facing suppression. A dense $z = 0$ matter-power check gives a top-hat-windowed amplitude ratio $\sigma_{8,\text{model}}/\sigma_{8,\text{ref}} \simeq 0.949$, confirming the geometric proxy at the 2×10^{-3} level. With a Planck 2018 reference amplitude $S_{8,\text{ref}} = 0.831$, the compressed amplitude proxy is $S_8 \simeq 0.787$, in the range of published low-redshift weak-lensing constraints. The calculation is not a gauge-complete Boltzmann closure and does not constitute a DES/KiDS/HSC/Planck likelihood analysis.

I. INTRODUCTION

Late-time suppression of matter clustering is a recurring target for models addressing the low-redshift growth amplitude inferred from weak lensing, galaxy clustering, and redshift-space distortions [1–9]. A useful mechanism should satisfy several requirements. It should reduce the effective matter clustering amplitude at late times, avoid excessive distortions of the background distance ladder, remain screened or weakly active at early times, and not introduce uncontrolled scale dependence into the matter power spectrum.

In this work we study a late-time mechanism that we call the *Selection-Fold Dark Drain*. Its defining chain is

$$\begin{aligned} U_{\text{sel}}(u; \Omega_c) &\implies u \text{ fold} \implies x \text{ entrainment} \implies m_c(x) \\ &\implies \Omega_c \text{ drain} \implies P(k), S_8 \downarrow. \end{aligned} \tag{1}$$

Here $u(N)$ is a selection-branch coordinate, $x(N) = \theta/M_{\text{Pl}}$ is a physical dark scalar, and

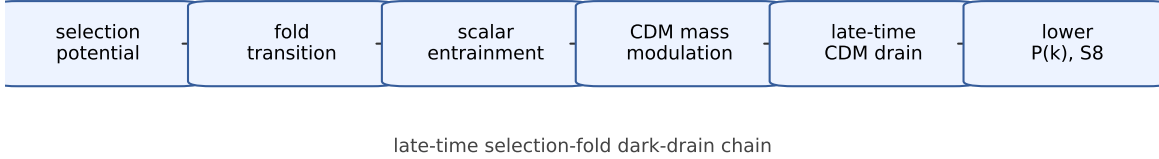


FIG. 1. Selection-fold dark-drain mechanism. A fold transition in the selection branch coordinate entrains a physical scalar, modulates the cold-dark-matter mass, drains Ω_c at late times, and suppresses the linear matter-power amplitude.

$N = \ln a$ is the e-fold time.

This paper focuses on a benchmark implementation rather than a cosmological data analysis. Its purpose is to establish that the selection-fold mechanism can be implemented consistently at the background level, that a controlled effective perturbation bridge can imprint a stable late-time matter-power suppression into CLASS output without large scale-dependent tilt, and that a Newtonian-gauge CLASS implementation can register $(\delta u, \delta x)$, verify the matter-source insertion point, preserve baseline parity, and admit a bounded final replay while remaining short of gauge-complete closure.

The presentation follows the calculation. We first define the fold-driven background model, then construct the CLASS background bridge and the source-layer matter-power insertion. We next replace the effective source by quasistatic and dynamic response variants, test the coupled growth response, and finally evaluate the Newtonian-gauge CLASS branch and its S_8 -scale interpretation. Throughout, we distinguish implementation tests from physical gauge-complete closure and from likelihood-level cosmological inference.

The main results are: (i) a fold-corrected CLASS background bridge with sub-percent expansion residuals and a derivative-level consistency check at 7.68×10^{-10} ; (ii) identification and validation of the final CLASS matter-source insertion chain; (iii) response-derived and dynamic source replacements with controlled matter-power tilt; (iv) a Newtonian-gauge CLASS benchmark reaching $P_{\text{geo}} \simeq 0.8971$ with tilt $\simeq 1.61\%$; and (v) an S_8 -facing amplitude proxy of approximately 5.3%.

The theoretical role of the selection potential in this work is deliberately limited. We do not assume a complete microscopic theory of branch selection. Instead, U_{sel} is used as

a local effective potential obtained after coarse graining over faster boundary, readout, and fluctuation sectors. A useful operational interpretation is that U_{sel} represents the local continuum limit of a branch-selection cost that favors dynamically stable, low-leakage branches with resolvable coarse graining and environmental support. The cosmological model then uses the lowest-order fold normal form of this effective potential, with the CDM abundance acting as the control variable. This places the construction in the standard logic of effective field theory: the detailed ultraviolet origin is not fixed, but the allowed low-energy branch dynamics, response scales, and stability diagnostics are made explicit and tested in CLASS.

II. SELECTION-POTENTIAL MOTIVATION

We use the selection potential as an effective local free-energy functional for slow branch variables. Let $X^A(\lambda)$ denote the slow collective coordinates that remain after coarse graining, and let b, Y, ζ denote boundary, readout, and fast fluctuation sectors. Integrating out these sectors defines an effective action for the slow variables. The term “selection potential” refers only to the local, low-derivative part of this effective action. This viewpoint is analogous in spirit to the use of coarse-grained free energies and local effective actions in phase-transition and effective-field-theory settings [10–13]. Formally, the path weight is

$$P[X] \propto \int \mathcal{D}b \mathcal{D}Y \mathcal{D}\zeta \exp \left[-\frac{1}{\varepsilon} S_{\text{tot}}[X, b, Y, \zeta] \right], \quad (2)$$

and we define

$$\Gamma_{\text{eff}}[X] = -\varepsilon \log P[X]. \quad (3)$$

In a low-derivative, local, weak-memory approximation,

$$\Gamma_{\text{eff}}[X] = \int d\lambda \left[\frac{1}{2} G_{AB}(X) \dot{X}^A \dot{X}^B + U_{\text{sel}}(X) \right] + \Delta\Gamma_{\text{nonlocal}}. \quad (4)$$

A generic local selection potential may contain terms of the schematic form

$$\begin{aligned} U_{\text{sel}}(X) = & U_0(X) + \lambda_b \Phi(X)^2 + \lambda_g (\mathcal{K}(X) - \mathcal{K}_\star)^2 \\ & + \frac{1}{2} (O(X) - R)^T C^{-1} (O(X) - R) - \tau_R \log \det (1 + \ell^2 \mathfrak{F}(X)) \\ & + \frac{\varepsilon}{2} \log \det' \mathcal{K}_{\text{fluc}}(X) + \Lambda_{\text{neg}} N_- (\mathcal{K}_{\text{fluc}}) + \cdots, \end{aligned} \quad (5)$$

where $\mathfrak{F}_{AB} = \partial_A O_i C_{ij}^{-1} \partial_B O_j$. The terms in Eq. (5) represent possible local costs and rewards generated by coarse graining, including boundary anchoring, readout mismatch, fluctuation stability, and negative-mode penalties.

A complementary operational motivation comes from branch-model selection in open systems, decoherence theory, Quantum Darwinism, and minimum-description length model selection [14–18]. In such a view, stability, leakage, coarse-graining cost, and environmental support contribute to an effective branch cost. We use this only to motivate the signs and roles of local terms in U_{sel} , not as an additional dynamical law.

Equation (5) should be read as a template for the possible local terms generated by coarse graining, not as a claim that every term is independently measured in the present cosmological benchmark. The dark-drain model below uses only the minimal one-dimensional fold sector and its two-field response completion. The remaining terms provide a bookkeeping language for stability, readout alignment, fluctuation penalties, and coarse-graining cost.

A useful bridge to effective field theory is obtained by expanding around a stable selected branch $X_\star$. The Hessian

$$H_{AB} = \nabla_A \nabla_B U_{\text{sel}}(X_\star) \quad (6)$$

gives the effective mass spectrum through the generalized eigenvalue problem

$$H_{AB} v_n^B = m_n^2 G_{AB}^\star v_n^B. \quad (7)$$

Thus selected branches are not merely kinematic labels. Their local curvature defines the effective field spectrum. This is the only aspect of the broader selection picture used later in the perturbation analysis: the response sector is organized by the curvature and stability of the selected branch, while the CLASS benchmark tests whether the resulting effective source can be transmitted to matter power without large scale dependence.

III. TWO-FIELD SELECTION-FOLD DARK-DRAIN MODEL

A. Fold potential

The active selection direction is modeled by the minimal normal form of a one-dimensional fold transition. Near such a transition, the full high-dimensional selection landscape is assumed to reduce to one soft branch coordinate u , while the remaining directions are treated as stable or integrated out. The quartic-minus-quadratic potential below is therefore used as an effective fold normal form, not as a microscopic ultraviolet potential.

The cosmological mechanism uses a one-dimensional selection branch coordinate $u(N)$ with potential

$$U_{\text{sel}}(u; \Omega_c) = \frac{1}{4}u^4 - \frac{1}{2}u^2 + J_{\text{eff}}(\Omega_c)u. \quad (8)$$

The control variable is

$$J_{\text{eff}}(\Omega_c) = -J_0 \tanh \left[\frac{1 - \Omega_c/\Omega_{\text{tr}}}{\sigma_J} \right]. \quad (9)$$

The fold critical value of the cubic branch equation is

$$J_{\text{crit}} = \frac{2}{3\sqrt{3}}. \quad (10)$$

A key convention identified in the CLASS bridge is that the threshold is not the naive value $\Omega_{c,\text{ref}}a_{\text{tr}}^{-3}$, but rather the fold-corrected value. With the sign convention in Eq. (9), the benchmark transition is defined by the branch crossing $J_{\text{eff}} = -J_{\text{crit}}$ at $a = a_{\text{tr}}$. This gives

$$\Omega_{\text{tr}} = \frac{\Omega_{c,\text{ref}}a_{\text{tr}}^{-3}}{1 - \sigma_J \text{atanh}(J_{\text{crit}}/J_0)}. \quad (11)$$

This correction is essential for reproducing the early high-density shoulder of the drain source.

B. Physical scalar and dark-matter drain

Throughout this subsection, $\Omega_i(N)$ denotes the density component normalized to the present critical density, not the instantaneous fractional density divided by $E^2(N)$.

The physical scalar is

$$x(N) = \theta/M_{\text{Pl}}. \quad (12)$$

It couples to the cold-dark-matter mass through

$$m_c(x) = m_{c0} \exp[-\beta_x(x - x_{\text{ini}})]. \quad (13)$$

The cumulative drain is

$$d(N) = \beta_x[x(N) - x_{\text{ini}}], \quad (14)$$

and the cold-dark-matter density evolves as

$$\Omega_c(N) = \Omega_{c,\text{ref}} \exp[-3N - d(N)]. \quad (15)$$

We define the raw drain rate as

$$\Gamma_{\text{raw}} = \beta_x x'. \quad (16)$$

Prescription	Drain behavior	Suppression response	Interpretation
Zero floor $\max(\Gamma_{\text{raw}}, 0)$	fiducial late-time drain	fiducial output	used in main runs
Signed Γ_{raw}	bounded; no runaway reversal	consistent qualitative behavior	not a hard-floor artifact
Smooth zero floor	stable late-time drain	consistent qualitative behavior	smooth replacement check
Naive softplus	high- z leakage	rejected	not used in benchmark

TABLE I. Drain-prescription robustness summary. The comparison records which transfer prescriptions were retained or rejected in the benchmark construction. The zero-floor prescription is used in the main runs; the signed and smooth zero-floor variants provide prescription-level sanity checks, while the naive softplus variant is excluded because it introduces unwanted high-redshift leakage.

In the numerical bridge we use the zero-floor transfer prescription

$$\Gamma_{\text{drain}} = \max(\Gamma_{\text{raw}}, 0). \quad (17)$$

The cold-dark-matter evolution and the transfer equations below use this Γ_{drain} :

$$\Omega'_c = -(3 + \Gamma_{\text{drain}})\Omega_c. \quad (18)$$

The drained energy is routed into a selection-like dark component and a fast-redshifting component:

$$\Omega'_{\text{sel}} = -3(1 + w_{\text{sel}})\Omega_{\text{sel}} + \epsilon_{\text{DE}}\Gamma_{\text{drain}}\Omega_c, \quad (19)$$

$$\Omega'_{\text{red}} = -3(1 + w_{\text{red}})\Omega_{\text{red}} + (1 - \epsilon_{\text{DE}})\Gamma_{\text{drain}}\Omega_c, \quad (20)$$

with

$$w_{\text{sel}} = -1, \quad w_{\text{red}} = 1. \quad (21)$$

The drain prescription checks used to define the fiducial transfer rule are summarized in Table I.

C. Two-field dynamics

The full two-field potential is

$$U_{\text{total}}(u, x; \Omega_c) = U_{\text{sel}}(u; \Omega_c) + \frac{1}{2}m_L^2(x - u)^2 + \frac{1}{2}m_x^2(x - x_{\text{ini}})^2. \quad (22)$$

The first-order relaxation dynamics used in the benchmark are

$$u' = -\Gamma_u \left[u^3 - u + J_{\text{eff}} - m_L^2(x - u) \right], \quad (23)$$

$$x' = -\Gamma_x \left[m_L^2(x - u) + m_x^2(x - x_{\text{ini}}) \right]. \quad (24)$$

A direct fit to the standalone background table confirms the benchmark values

$$\Gamma_u \simeq 12, \quad \Gamma_x \simeq 200, \quad m_L^2 \simeq 2, \quad m_x^2 \simeq 0. \quad (25)$$

D. Relation to interacting dark-sector benchmarks

The present construction is related to interacting dark-energy and coupled-dark-matter scenarios, in which energy exchange or scalar-mediated forces modify the late-time growth history [19–24]. The distinguishing feature here is that the interaction is not introduced as a constant coupling or as a prescribed phenomenological transfer function. It is triggered by a fold-like branch transition controlled by the CDM abundance, which then entrains the scalar controlling the dark-matter mass. In this sense the model is best viewed as a CLASS benchmark for a transition-triggered dark-sector drain, rather than as a completed likelihood-level interacting dark-energy model. Table II summarizes which quantities are physical dark-sector inputs, effective bridge parameters, numerical normalizations, or diagnostic-only coordinates.

IV. STANDALONE BENCHMARK

The main standalone benchmark uses

$$\begin{aligned} J_0 &= 0.8, & \epsilon_{\text{DE}} &= 0.16, & \Gamma_u &= 12, \\ \Gamma_x &= 200, & m_L^2 &= 2.0, & m_x^2 &= 0. \end{aligned} \quad (26)$$

The original standalone bridge gives

$$d_{\text{transfer,final}} \simeq 0.069052, \quad d_{\text{field,final}} \simeq 0.069059, \quad (27)$$

$$\Omega_{\text{c0}}^{\text{eff}} \simeq 0.242652, \quad \Omega_{m0}^{\text{eff}} \simeq 0.292652, \quad (28)$$

$$\Omega_{\text{sel},0} \simeq 0.043877, \quad \Omega_{\text{red},0} \simeq 1.536 \times 10^{-3}, \quad (29)$$

$$\Omega_{\Lambda,\text{model}} \simeq 0.661845, \quad \max |x - u| \simeq 0.02849. \quad (30)$$

Parameter	Role	Fixed by	Status
J_0	fold amplitude	benchmark choice	dark-sector benchmark input
a_{tr}	activation epoch	background source alignment	dark-sector benchmark input
σ_J	transition width	background source alignment	dark-sector benchmark input
β_x	CDM mass-drain strength	background drain normalization	dark-sector benchmark input
ϵ_{DE}	drain routing fraction	standalone benchmark	phenomenological input
m_L^2, m_x^2	two-field locking and scalar stiffness	background replay	model input
Γ_u, Γ_x	relaxation rates	background replay	effective dynamics
C_{drain}	effective source amplitude	source bridge scan	bridge parameter
C_x	response-source amplitude	response bridge scan	bridge parameter
p	response gate power	response bridge scan	phenomenological gate
g_{grow}	source-to-growth projection gain	coupled replay calibration	bridge calibration
Γ_{mult}	response relaxation normalization	local envelope scan	numerical normalization
X_{norm}	online response normalization	bounded CLASS replay	numerical normalization
Λ_{met}	metric-source diagnostic coordinate	finite metric audit	diagnostic only

TABLE II. Parameter roles in the benchmark. The table separates physical dark-sector inputs from effective bridge parameters, numerical normalization choices, and diagnostic-only quantities.

The effective growth replay gives

$$S_8^{\text{proxy}} \simeq 0.94710, 0.94797, 0.95056, 0.95489, 0.96097 \quad (31)$$

for

$$\beta_{\text{eff}} = 0, 0.05, 0.10, 0.15, 0.20. \quad (32)$$

This demonstrates that the benchmark remains in a few-percent suppression regime under activated fifth-force-like corrections.

V. CLASS BACKGROUND BRIDGE

A. Bridge implementation

We implemented a background-only bridge in CLASS [25], in the same Boltzmann-solver ecosystem as CAMB and its modified-gravity extensions [26–28]. The bridge evolves u , x ,

Ω_c , Ω_{sel} , Ω_{red} , and d_{transfer} , and outputs diagnostic columns including

$$u, x, \Gamma_{\text{drain}}, \Omega_c, \Omega_{\text{sel}}, \Omega_{\text{red}}, d_{\text{transfer}}, \quad (33)$$

as well as the reconstructed Friedmann branch

$$E_{\text{Mo}}^2(N) = \Omega_b(N) + \Omega_r(N) + \Omega_c(N) + \Omega_{\text{sel}}(N) + \Omega_{\text{red}}(N) + \Omega_{\Lambda}^{\text{Mo}}. \quad (34)$$

Here the subscript “Mo” labels the reconstructed model branch used as the background handoff target. In the latest audit branch we further exposed internal RHS diagnostics

$$J_{\text{eff}}, dU_d u, dU_d x, u'_{\text{rhs}}, x'_{\text{rhs}}, \quad (35)$$

and echo columns for

$$\Omega'_c, \Omega'_{\text{sel}}, \Omega'_{\text{red}}, d'_{\text{transfer}}, \quad (36)$$

to verify that integrated derivatives and printed diagnostics are generated from the same code path.

A gated handoff was introduced:

$$H_{\text{CLASS}} \leftarrow H_{\text{Mo}} \quad (37)$$

when the Mo bridge is active. This lowers the main $H(z)$ mismatch relative to the standalone target.

B. Source-shape alignment

Before implementing the fold-corrected threshold, the CLASS bridge missed the early high-density shoulder of the drain source. The key diagnostic was the source budget

$$I_{\text{source}} = \int \Gamma_{\text{drain}}(N) \Omega_c(N) dN. \quad (38)$$

The standalone bridge has

$$I_{\text{source}}^{\text{target}} \simeq 0.274232, \quad \langle \Omega_c \rangle_{\Gamma} \simeq 3.971. \quad (39)$$

After implementing Eq. (11) and locally refining $(a_{\text{tr}}, \sigma_J, \beta_x)$, we find a source-shape bridge point

$$a_{\text{tr}} = 0.335, \quad \sigma_J = 0.78, \quad \beta_x = 0.0274125, \quad (40)$$

with

$$I_{\text{source}} \simeq 0.2680, \quad z_{\text{peak}} \simeq 0.9024, \quad \text{area}(N < -1.2) \simeq 0.0958. \quad (41)$$

This is close to the standalone target

$$I_{\text{source}}^{\text{target}} \simeq 0.2742, \quad z_{\text{peak}}^{\text{target}} \simeq 0.909, \quad \text{area}_{N < -1.2}^{\text{target}} \simeq 0.104. \quad (42)$$

C. Development audits and threshold convention

Several implementation audits were used to isolate the source of the early background mismatch. Start-time tests and derivative-variable checks ruled out initialization and $d/d \ln a$ convention errors. A single-source RHS helper was then introduced so that the background integrator and diagnostic output shared the same Mo-sector derivative path, reducing RHS parity residuals to the 10^{-5} – 10^{-4} level.

The remaining source-budget discrepancy was traced to the control mapping in $J_{\text{eff}}(\Omega_c)$. Direct inversion of the benchmark control trajectory showed that the transition threshold must use the fold-corrected convention in Eq. (11), rather than the naive $\Omega_{c,\text{ref}} a_{\text{tr}}^{-3}$ value. After this correction, the CLASS bridge recovers the early source shoulder and provides the benchmark background used in the later perturbation and response stages. The detailed audit trail is retained in the reproducibility manifest rather than repeated in the main text.

D. Background closure

The main consistency checks for the benchmark background are summarized in Table III.

The benchmark candidate is evaluated with the reconstructed-Friedmann handoff enabled and with bridge parameters

$$a_{\text{tr}} = 0.335, \quad \sigma_J = 0.78, \quad \beta_x = 0.0274125. \quad (43)$$

In addition, a derivative self-consistency audit gives

$$\max \left| \frac{d \ln H}{dN} - \frac{d \ln E_{\text{Mo}}}{dN} \right| \simeq 7.68 \times 10^{-10}, \quad (44)$$

showing that the handed-off H column and the reconstructed Friedmann branch are numerically indistinguishable at the derivative level.

Object	Test	Result	Interpretation
$H(z)$	relative residual vs reconstructed branch	5.56×10^{-4}	below 10^{-3}
$I(z)$	line-of-sight integral residual, $z > 0$	4.02×10^{-4}	below 10^{-3}
$d \ln H / dN$	derivative residual vs reconstructed branch	7.68×10^{-10}	derivative-level agreement
Geometry	optional CLASS synchronization columns	3×10^{-3} level	secondary check

TABLE III. Summary of the CLASS background bridge consistency checks. The main expansion and line-of-sight integral are below the 10^{-3} level, while optional geometry and time columns provide a secondary synchronization check.

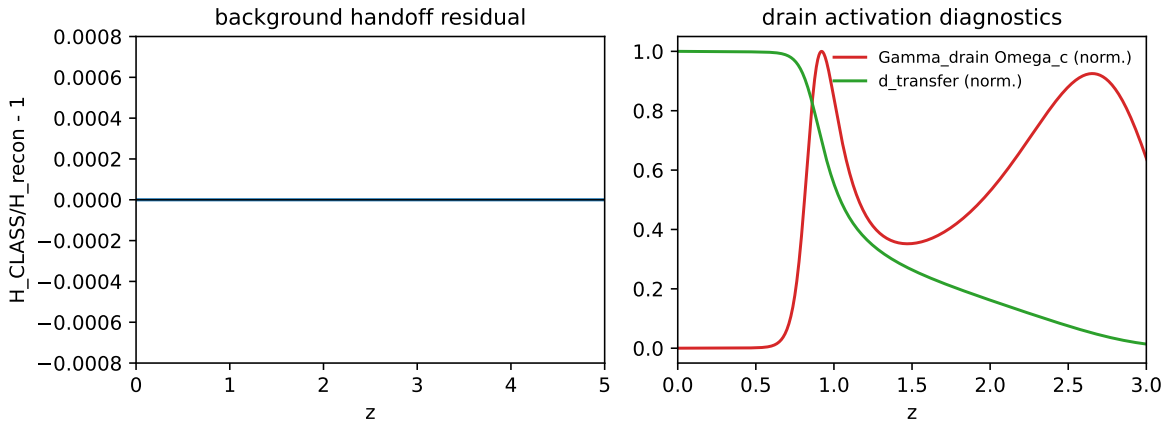


FIG. 2. Background bridge diagnostics. The left panel shows the relative handoff residual between the CLASS background and the reconstructed Friedmann branch. The right panel shows normalized drain-source and cumulative-transfer diagnostics, illustrating the late-time activation window used by the perturbation benchmarks.

For optional geometry/time columns, we use corrected geometric relations $D_C \propto I$, $D_A \propto I/(1+z)$, $D_L \propto (1+z)I$, and an affine fit $\eta \simeq A + BI$, with low- z and ultra-high- z tails treated separately in diagnostics. This confirms that the central bridge requirement is the (H, I) pair, while built-in geometry/time columns are a secondary synchronization layer.

VI. EFFECTIVE PERTURBATION BRIDGE

A. Standalone effective replay

Before modifying the CLASS perturbation output, we replayed the effective subhorizon growth equations using the benchmark CLASS background. The two-fluid replay evolves CDM and baryon growth variables with an activated fifth-force-like correction

$$\mu_x(k, N) = 1 + 2\beta_{\text{eff}}^2 A_x(N) \frac{k^2}{k^2 + a^2 m_{\text{eff}}^2(N)}, \quad (45)$$

where

$$m_{\text{eff}}^2/H_0^2 = M_0^2 + M_1^2 A_x(N). \quad (46)$$

Under the internal convention $\Omega_{m0}^{\Lambda\text{CDM}} = 0.31$, all 32 mixed and CDM-only replay branches satisfy the stated stability criteria, with

$$S_8^{\text{proxy}} \simeq 0.95064\text{--}0.95709, \quad |\text{tilt}| \sim 0.002\%. \quad (47)$$

A stress convention $\Omega_{m0}^{\Lambda\text{CDM}} = 0.315$ initially fails the hard $S_8 \geq 0.94$ lower cut in some branches, but passes under $S_8 \geq 0.935$. This identifies the effect as a normalization stress sensitivity, not a scale-dependent instability. For completeness, we performed an explicit convention calibration replay. The official branch with $\Omega_{m0}^{\Lambda\text{CDM}} = 0.31$ satisfies the criteria across all 32 tested branches. A stress branch including $\Omega_{m0}^{\Lambda\text{CDM}} = 0.315$ also satisfies the relaxed criterion $S_8 \geq 0.935$ across all 64 tested branches. Therefore the previous partial-fail pattern at fixed $S_8 \geq 0.94$ is a convention-threshold effect, not a dynamical instability.

B. CLASS source-chain tracing

A direct Newtonian-gauge Euler hook was first tested:

$$\dot{\theta}_c = -(1 + \Gamma_N) \mathcal{H} \theta_c + \mu_{\text{eff}} \text{metric_euler}. \quad (48)$$

Although the hook executed, it did not affect the final $P(k)$ output. We therefore traced the CLASS matter-power source chain and verified that the Fourier read-side source array is the final matter-power insertion point. To exclude stale-output artifacts, we repeatedly regenerated the off/on comparison products before each test. A read-side source-response

control that scaled the relevant matter source by a factor of 0.5 produced the expected

$$P_{\text{on}}(k)/P_{\text{off}}(k) = 0.25 \quad (49)$$

across the sampled k values. This identifies the Fourier read-side chain as the final mPk insertion point.

We then performed write/read memory-instance tracing at matched grid points. For matched time and wavenumber samples, the stored perturbation-source value and the Fourier-read value agree pointwise. Hence the effective chain from the perturbation source array to final mPk is directly verified.

Finally, a pointwise source-bias sanity test applied only on a restricted low- k grid subset gives

$$P_{\text{on}}/P_{\text{off}} = 0.25 \quad (50)$$

for low- k sampled points while higher- k sampled points remain unchanged. This resolves the earlier apparent contradiction from broad but non-pointwise diagnostics and confirms that source-layer perturbation insertions are operational.

Pointwise source-bias tests confirmed that modifying the matter perturbation source propagates exactly into the final mPk output. This motivated an intermediate effective source-layer bridge.

C. Gauge-split perturbation access

Before any physical perturbation insertion, we performed a hook-only reconnaissance of the perturbation equations. The audit confirms that the CDM perturbation equations are gauge-split in the standard CLASS form:

$$\delta'_c = -(\theta_c + \text{metric_continuity}), \quad \theta'_c = -\mathcal{H}\theta_c + \text{metric_euler} \quad (\text{newtonian}), \quad (51)$$

$$\delta'_c = -\text{metric_continuity}, \quad \theta_c \equiv 0 \quad (\text{synchronous}). \quad (52)$$

The velocity index for CDM is defined only in Newtonian gauge, while synchronous gauge uses the gauge condition $\theta_c = 0$. This reconnaissance fixes the gauge scope of the effective bridge: the present implementation is Newtonian-first, while a synchronous-consistent closure is left for future work.

D. Effective source-layer bridge

We define an effective source multiplier

$$\delta_m^{\text{source}} \rightarrow S_{\text{eff}}(N, k) \delta_m^{\text{source}}. \quad (53)$$

The minimal form used in this work is

$$S_{\text{eff}}(N, k) = 1 - A_{\text{drain}} A_x(N) + A_\mu A_x(N) \frac{k^2}{k^2 + a^2 H_0^2 M_0^2}. \quad (54)$$

Here

$$A_x(N) = \frac{d_{\text{transfer}}(N)}{d_{\text{transfer}}(0)}. \quad (55)$$

The implementation also preserves a constant-source fallback.

A diagnostic constant multiplier $S_{\text{eff}} = 0.98$ gives

$$P(k)_{\text{on}}/P(k)_{\text{off}} = (0.98)^2 = 0.9604 \quad (56)$$

at all tested k , confirming that the source-layer insertion is stable.

VII. MATTER-POWER RESULTS

For a set of sampled wavenumbers k_j , we summarize the matter-power ratio by

$$P_{\text{geo}} = \exp \left[\frac{1}{N_k} \sum_{j=1}^{N_k} \ln \left(\frac{P_{\text{on}}(k_j)}{P_{\text{ref}}(k_j)} \right) \right]. \quad (57)$$

We denote by P_{min} and P_{max} the minimum and maximum of the same ratio over the sampled k range. The quoted residual scale dependence is

$$\text{tilt} = 100 \max_j \left| \frac{P_{\text{on}}(k_j)/P_{\text{ref}}(k_j)}{P_{\text{geo}}} - 1 \right| \%. \quad (58)$$

A. Calibration scan

We scanned

$$A_{\text{drain}} = 0.015, 0.020, 0.025, 0.030, \quad (59)$$

$$A_\mu = 0, 0.003, 0.005, 0.008, \quad (60)$$

$$M_0 = 10, 30, 100, \quad (61)$$

case	A_{drain}	A_{μ}	M_0	$P_{0.01}/P_{\text{off}}$	$P_{0.03}/P_{\text{off}}$	$P_{0.10}/P_{\text{off}}$	$P_{0.30}/P_{\text{off}}$	$\min_{z=0\sim 2} P/P_{\text{off}}$	$\max_{z=0\sim 2} P/P_{\text{off}}$	$\max \text{tilt} $ (%)
conservative	0.020	0.000	10	0.9604	0.9604	0.9604	0.9604	0.9604	0.9935	≈ 0
mild k -dependent	0.025	0.005	10	0.9594	0.9603	0.9604	0.9604	0.9594	0.9935	0.01
boundary	0.030	0.000	10	0.9409	0.9409	0.9409	0.9409	0.9409	0.9903	≈ 0

TABLE IV. Representative source-layer effective bridge points. The conservative benchmark gives scale-independent suppression at $z = 0$, the mild benchmark introduces only tiny k -dependence, and the boundary point illustrates low-redshift amplitude over-suppression.

with $M_1 = 0$. The scan contains 48 points; 43 satisfy the baseline diagnostic criteria and 13 also satisfy the stricter ideal-window criteria. The best conservative points have

$$A_{\text{drain}} = 0.020, \quad A_{\mu} = 0, \quad (62)$$

with

$$P(k)/P_{\text{off}} = 0.9604 \quad (63)$$

uniformly across

$$k = 0.01, 0.03, 0.10, 0.30 \, h \, \text{Mpc}^{-1}. \quad (64)$$

A representative mild k -dependent point is

$$A_{\text{drain}} = 0.025, \quad A_{\mu} = 0.005, \quad M_0 = 10, \quad (65)$$

which gives

$$P(k)/P_{\text{off}} \simeq 0.9594\text{--}0.9604 \quad (66)$$

and a tilt of only

$$0.0127\%. \quad (67)$$

B. Representative points

We use three representative points, summarized in Table IV.

C. Redshift stability

The representative points were tested at

$$z = 0, 0.5, 1, 2. \quad (68)$$

The conservative point gives

$$P(k)/P_{\text{off}} = 0.9604 \quad (z = 0), \quad (69)$$

$$\simeq 0.9605 \quad (z = 0.5), \quad (70)$$

$$\simeq 0.9780 \quad (z = 1), \quad (71)$$

$$\simeq 0.9935 \quad (z = 2), \quad (72)$$

with negligible tilt at all tested redshifts.

The mild k -dependent point gives

$$P(k)/P_{\text{off}} \simeq 0.9594\text{--}0.9604 \quad (73)$$

at $z = 0$, with maximum tilt

$$0.0127\%. \quad (74)$$

The tilt decreases to

$$2.3 \times 10^{-4}\% \quad (75)$$

by $z = 2$.

The boundary point gives

$$P(k)/P_{\text{off}} \simeq 0.9409 \quad (76)$$

at $z = 0$, rising to

$$0.9903 \quad (77)$$

by $z = 2$. It is therefore a low-redshift amplitude over-suppression boundary, not a large-tilt boundary.

D. Quasistatic response-derived source bridge

To move beyond a purely hand-tuned source multiplier, we constructed a response-background dataset from the benchmark CLASS background output and analyzed a quasistatic two-field response kernel. The background Hessian

$$H = \begin{pmatrix} H_{uu} & H_{ux} \\ H_{xu} & H_{xx} \end{pmatrix} \quad (78)$$

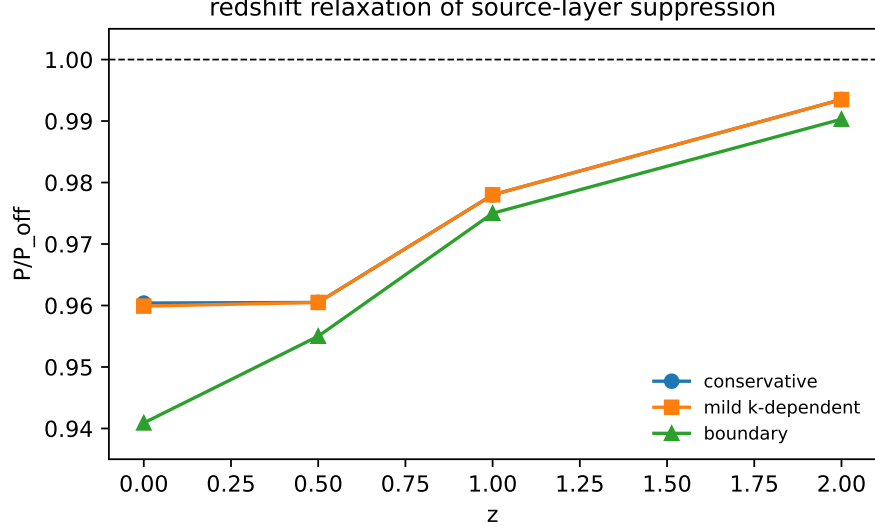


FIG. 3. Redshift relaxation of representative source-layer matter-power suppression. The conservative and mild benchmarks give smooth late-time suppression and relax toward unity by $z \simeq 2$; the boundary point is an amplitude boundary rather than a scale-tilt pathology.

shows a short fold-soft interval with negative determinant/eigenvalue support. Quantitatively, the negative-eigenvalue fraction is

$$f_{\lambda_{\min}(H)<0} \simeq 0.0078, \quad (79)$$

localized around

$$z \simeq 0.91\text{--}1.46. \quad (80)$$

This soft interval is physically expected near a fold transition.

We then considered the finite- k response kernel

$$K = H + Q_k e_x e_x, \quad Q_k \propto \frac{k^2}{a^2 H_0^2}, \quad (81)$$

with regularized/damped variants as controls. Over the tested observational window

$$k = 0.01, 0.03, 0.10, 0.30 \, h \, \text{Mpc}^{-1}, \quad (82)$$

the kernel is lifted to non-negative minimum eigenvalues, i.e. the fold soft mode is stabilized by finite-gradient stiffness. Damping further reduces stiffness modestly but does not change the qualitative conclusion. In this sense, the data favor a “soft-but-regularized” picture: the bare fold sector is locally soft, but the finite-gradient term keeps the tested response kernel

case	key parameters	$f_{\text{resp}}(z=0)$	$\min_{z,k} P/P_{\text{off}}$	$\max \text{tilt} $ (%)
source-layer conservative	$A_{\text{drain}} = 0.020, A_{\mu} = 0$	0	0.9604	0.0000
source-layer mild k -dependent	$A_{\text{drain}} = 0.025, A_{\mu} = 0.005, M_0 = 10$	0	0.9594	0.0127
response-derived robust	$p = 3.25, C_{\text{drain}} = 0.014, C_x = 0.015$	0.0619	0.9451	0.3100
response-derived stress	$p = 3.25, C_{\text{drain}} = 0.014, C_x = 0.017$	0.0696	0.9415	0.3514

TABLE V. Source-layer effective bridge versus gated response-derived bridge.

invertible over the observational k window. Numerically, the bare Hessian reaches a modest peak condition number $\sim 7.1 \times 10^3$, while the response kernel is non-negative but extremely stiff (typical condition numbers $\sim 10^{30}$ – 10^{33}), motivating normalized-response and clipping safeguards in the prototype scan.

Using the response-derived ansatz

$$S_{\text{phys}}(N, k) = 1 - C_{\text{drain}} A_x(N) + C_x \hat{R}_x(N, k), \quad (83)$$

with normalized response $\hat{R}_x$ and clipping safeguards, we ran an 84-point standalone prototype scan:

$$4 (C_{\text{drain}}) \times 7 (C_x) \times 3 (\text{kernel}) = 84. \quad (84)$$

The scan finds broad stable regions. For each kernel branch, 25/28 points satisfy the baseline criteria and 9/28 also satisfy the stricter ideal-window criteria, with the best point at

$$C_{\text{drain}} = 0.020, \quad C_x = 0, \quad (85)$$

recovering

$$P_{\text{ratio}}(z=0) \simeq 0.9604 \quad (86)$$

and negligible tilt, with zero clipping in top rows. This supports the interpretation that the effective source-layer bridge can be promoted to a response-derived prototype without introducing large k -shape distortions.

E. Source-layer and response-bridge comparison

To consolidate the development chain, we assembled a direct comparison between the source-layer bridge and the response-derived bridge, summarized in Table V.

The source-layer bridge establishes that an effective correction can stably imprint the target suppression into CLASS matter power without large scale-dependent tilt. The response-derived bridge then introduces a correction from the two-field fold kernel with a late-time activation gate. The robust gated benchmark

$$G_R = A_x^{3.25}, \quad C_{\text{drain}} = 0.014, \quad C_x = 0.015 \quad (87)$$

gives

$$f_{\text{resp}}(z=0) \simeq 0.0619, \quad \min_{z,k} P/P_{\text{off}} \simeq 0.9451, \quad \text{tilt} \simeq 0.31\%, \quad (88)$$

with zero clipping. A higher-response stress point with $C_x = 0.017$ raises $f_{\text{resp}}(z=0)$ to ~ 0.0696 but lowers the all-redshift floor to ~ 0.9415 , indicating the onset of the safe-window boundary.

F. Dynamic-response calibration

Before modifying CLASS perturbation equations, we tested a standalone first-order dynamic prototype for

$$\eta_u = \frac{\delta u}{\delta_c}, \quad \eta_x = \frac{\delta x}{\delta_c}, \quad (89)$$

with damped response-kernel coefficients matched to the response-derived benchmark branch. For each sampled mode

$$k = 0.01, 0.03, 0.10, 0.30 \, h \, \text{Mpc}^{-1}, \quad (90)$$

we evolved the dynamic system on $N \in [-8, 0]$ and compared $\eta_x(N, k)$ against the quasistatic kernel response $R_x^{\text{QS}}(N, k)$.

The prototype is numerically stable for all tested modes and initializations. The dynamic response tracks the quasistatic shape with high correlation,

$$\text{corr}(\eta_x, R_x^{\text{QS}}) \simeq 0.993\text{--}0.997, \quad (91)$$

and moderate relative error,

$$\text{rms_rel}(\eta_x - R_x^{\text{QS}}) \simeq 0.097\text{--}0.129. \quad (92)$$

At late time ($z=0$), the amplitude ratio is close to unity,

$$\eta_x/R_x^{\text{QS}} \simeq 1.042, \quad (93)$$

while by $z = 1$ it increases to

$$\eta_x/R_x^{\text{QS}} \simeq 1.18. \quad (94)$$

This behavior supports the dynamic-response benchmark: the dynamic prototype reproduces the quasistatic response pattern with controlled residual bias, but does not yet constitute a full Boltzmann closure.

We then performed a relaxation calibration scan over

$$\Gamma_{\text{mult}} \in \{0.5, 1, 2, 5\}, \quad \alpha_{\text{damp}} \in \{1, 3, 10, 30\}, \quad (95)$$

and minimized a late-time weighted loss dominated by η_x/R_x^{QS} at $z = 0$. The best point

$$\Gamma_{\text{mult}} = 5, \quad \alpha_{\text{damp}} = 30 \quad (96)$$

gives $\eta_x/R_x^{\text{QS}}|_{z=0} \simeq 1.0086$, with mean correlation $\simeq 0.9997$ and mean relative error $\simeq 0.0268$ across the tested k values.

We then replaced the quasistatic response in the gated source bridge by the calibrated dynamic response,

$$S_{\text{phys}}^{\text{dyn}} = 1 - C_{\text{drain}}A_x + C_xA_x^p\hat{\eta}_x^{\text{dyn}}, \quad (97)$$

and replayed the representative benchmarks. For the robust point $(p, C_{\text{drain}}, C_x) = (3.25, 0.014, 0.015)$, we obtain

$$\begin{aligned} f_{\text{resp}}(z = 0) &\simeq 0.0613, \quad \min_{z,k} P/P_{\text{off}} \simeq 0.9453, \\ \text{tilt} &\simeq 0.314\%. \end{aligned} \quad (98)$$

This replay has zero clipping and only $\Delta \min(P/P_{\text{off}}) \sim -7.5 \times 10^{-5}$ and $\Delta f_{\text{resp}}(z = 0) \sim 10^{-4}$ relative to the quasistatic bridge.

Finally, we evaluated a local sensitivity envelope at fixed robust benchmark

$$\Gamma_{\text{mult}} \in \{3, 4, 5, 6, 8\}, \quad \alpha_{\text{damp}} \in \{20, 25, 30, 35, 40\}. \quad (99)$$

All 25/25 points satisfy the bridge safety criteria (response share floor, all- z power floor, tilt bound, and zero clipping). This indicates that the calibrated dynamic bridge is locally robust, not a fine-tuned single-point construction.

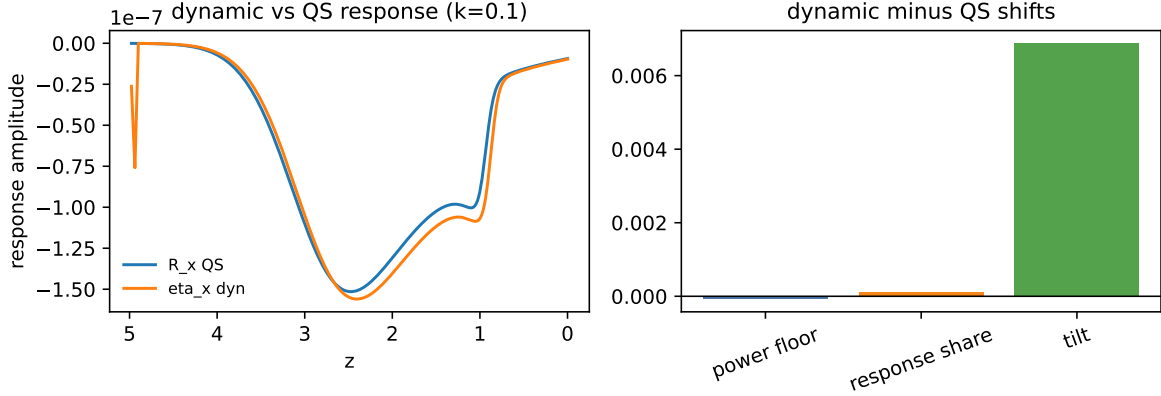


FIG. 4. Dynamic-response calibration. The left panel compares the quasistatic response proxy and the calibrated dynamic response for a representative mode. The right panel summarizes the small shifts induced by replacing the quasistatic response with the dynamic replay in the robust source benchmark.

G. Coupled growth-response solver and Picard feedback

To test whether the response-derived bridge survives growth integration, we introduced a standalone coupled growth-response solver with absolute variables $(D, V, \delta u, \delta x)$ and a frozen-response three-pass baseline:

$$D_{\text{off}} \rightarrow \widehat{X}^{(0)} \rightarrow D_{\text{on}}^{(1)}. \quad (100)$$

At direct source insertion $\mu_{\text{eff}} = S_{\text{dyn}}$, the coupled system is numerically stable but under-suppresses integrated growth relative to the dynamic source-level replay, indicating a non-trivial source-to-growth projection.

We therefore calibrated

$$\mu_{\text{eff}} = 1 + g_{\text{grow}}(S_{\text{dyn}} - 1). \quad (101)$$

For the robust benchmark $(p, C_{\text{drain}}, C_x) = (3.25, 0.014, 0.015)$ with $(\Gamma_{\text{mult}}, \alpha_{\text{damp}}) = (5, 30)$, the calibrated point

$$g_{\text{grow}} = 4 \quad (102)$$

gives

$$\min_{z,k} P/P_{\text{off}} \simeq 0.9511, \quad \text{tilt} \simeq 0.326\%, \quad \mu_{\text{eff,min}} \simeq 0.851, \quad \text{clip} = 0, \quad (103)$$

which places the coupled replay back into the dynamic-response suppression window. The corresponding calibrated-gain benchmark is summarized in Table VI.

g_{grow}	$f_{\text{resp}}(z=0)$	$\min_{z,k} P/P_{\text{off}}$	$\max_z \text{tilt} (\%)$	$\mu_{\text{eff,min}}$	clip	window
1	0.1245	0.9876	0.081	0.963	0.000	no
4	0.1245	0.9511	0.326	0.851	0.000	yes
5	0.1245	0.9392	0.408	0.814	0.000	no

TABLE VI. Calibrated gain benchmark for the robust coupled growth-response replay.

iteration	$\min_{z,k} P/P_{\text{off}}$	$\max_z \text{tilt} (\%)$	$f_{\text{resp}}(z=0)$	clip
iter1	0.9511	0.3260	0.12448	0.000
iter2	0.9513	0.3253	0.12336	0.000
iter3	0.9513	0.3253	0.12336	0.000
iter4	0.9513	0.3253	0.12336	0.000

TABLE VII. Picard convergence pilot for the calibrated robust coupled benchmark ($g_{\text{grow}} = 4$).

Iteration-to-iteration shifts remain at the 10^{-4} level in the power-floor metric, with zero clipping.

Finally, a one-step and mini-iteration Picard feedback pilot,

$$D_{\text{off}} \rightarrow \widehat{X}^{(0)} \rightarrow D_{\text{on}}^{(1)} \rightarrow \widehat{X}^{(1)} \rightarrow D_{\text{on}}^{(2)} \rightarrow \dots, \quad (104)$$

remains stable. In particular, the first update gives

$$\Delta P_{\text{min}} \simeq 1.45 \times 10^{-4}, \quad \Delta \text{tilt} \simeq 7.2 \times 10^{-4}\%, \quad \Delta f_{\text{resp}} \simeq -1.12 \times 10^{-3}, \quad (105)$$

with zero clipping, and subsequent updates remain at comparable or smaller deltas. This supports a feedback-stable calibrated coupled benchmark at the standalone level. The coupled response-share diagnostic is defined after source-to-growth projection and should not be numerically identified with the source-level response share. The Picard convergence table is summarized in Table VII.

H. Newtonian-gauge CLASS benchmark and metric diagnostics

After the standalone coupled replay, we add a Newtonian-first CLASS implementation layer for the response fields. The patch registers two perturbation variables $(\delta u, \delta x)$, evolves a relaxation-style response RHS in Newtonian gauge, and routes the resulting dynamic

source factor through the same final matter-source insertion point used in the source-chain tests. This implementation is designed as a Newtonian-gauge CLASS implementation and does not claim synchronous-gauge consistency or full Boltzmann closure.

The first acceptance test is a source-response control. A temporary half-amplitude source multiplier with $\mu_{\text{eff}} = 0.5$ gives

$$P_{\text{ctrl}}/P_{\text{ref}} = 0.25,$$

as expected for a source-level amplitude multiplier squared. A drain-only run with

$$S_{\text{dyn}} = 1 - C_{\text{drain}} A_x, \quad \mu_{\text{eff}} = 1 + g_{\text{grow}}(S_{\text{dyn}} - 1)$$

gives $P_{\text{drain-only}}/P_{\text{ref}} \simeq 0.8911$, confirming that the background activation A_x reaches the final mPk source insertion point. Runs with the closure disabled or with $C_{\text{drain}} = C_x = 0$ preserve baseline parity.

The dynamic response channel is then normalized online. At the bounded Newtonian-gauge benchmark,

$$\Gamma_{\text{mult}} = 0.005, \quad X_{\text{norm,mult}} = 7 \times 10^7,$$

the dynamic replay gives

$$P_{\text{geo}} \simeq 0.9039, \quad P_{\text{min}} \simeq 0.8896, \quad P_{\text{max}} \simeq 0.9351, \quad \text{tilt} \simeq 3.46\%.$$

A local refinement at

$$X_{\text{norm,mult}} = 1.5 \times 10^8$$

reduces the residual scale dependence to

$$P_{\text{geo}} \simeq 0.8971, \quad P_{\text{min}} \simeq 0.8896, \quad P_{\text{max}} \simeq 0.9116, \quad \text{tilt} \simeq 1.61\%,$$

with $\mu_{\text{eff}} \simeq 0.9440\text{--}0.9454$. A local envelope over $\Gamma_{\text{mult}} \in \{0.003, 0.005, 0.007\}$ and $X_{\text{norm,mult}} \in \{1.0, 1.2, 1.5, 1.8, 2.0\} \times 10^8$ gives 15/15 local-envelope successes and 6/15 target-2% successes. This defines the Newtonian-gauge CLASS response benchmark.

The metric channel is treated separately as a diagnostic numerical-coupling audit. We map the available Newtonian-gauge metric access points, wire metric-lambda hooks into the response helper, and verify that all zero-lambda metric configurations preserve the fiducial Newtonian-gauge benchmark. The metric-source scaling and placement ladder is retained as a diagnostic check: the `Hconf_inv2` rescaling restores matched-grid pointwise reversibility

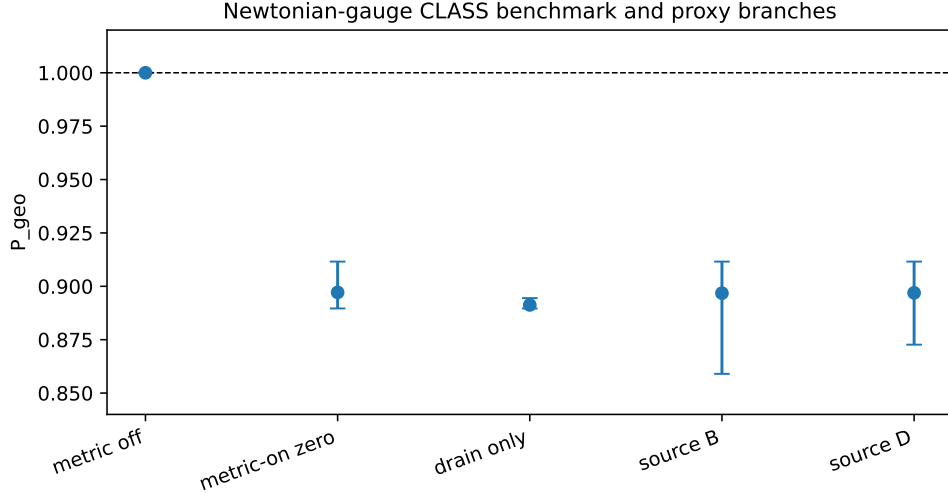


FIG. 5. Newtonian-gauge CLASS branch and executable source-family proxies. Points show the geometric mean matter-power ratio relative to the metric-off reference; bars show the corresponding $P_{\text{min}}-P_{\text{max}}$ range. The source-family branches are executable proxies and do not validate a physical nonzero metric-forcing model.

in the audited trace ladder, while inside-bracket placement is kept as the minimal-change default. This remains an implementation check and does not validate physical nonzero metric forcing.

The metric-on tests sharpen the boundary. With metric terms enabled, the metric-on-zero branch preserves the fiducial benchmark, and finite tiny/micro-lambda runs satisfy Tier-1 finite-output tests up to $|\lambda| = 10^{-5}$. However, no strict symmetric total bounded nonzero-metric window is found. An incremental audit relative to the metric-on-zero branch identifies a diagnostic micro-window at $|\lambda| = 10^{-10}$, motivating

$$\Lambda_{\text{met}} = \frac{|\lambda|}{10^{-10}}.$$

A normalized CLASS diagnostic test at $\Lambda_{\text{met}} \in \{0, 0.3, 1, 3\}$ satisfies the parity and finite-output gates, while the incremental gates pass only at $\Lambda_{\text{met}} = 1$. Thus Λ_{met} is useful as a normalized diagnostic coordinate, but not as a monotonic safety parameter or a physical coupling calibration.

I. S8-facing proxy interpretation

To connect the CLASS benchmarks to the low-redshift clustering target, we translate the geometric mean matter-power suppression into a simple amplitude proxy,

$$\frac{S_{8,\text{model}}}{S_{8,\text{ref}}} \simeq \sqrt{P_{\text{geo}}}.$$

This relation is used only as a scale translation. It is exact only for a scale-independent matter-power suppression. For a scale-dependent suppression, one would compute the windowed amplitude ratio

$$\frac{\sigma_{8,\text{model}}}{\sigma_{8,\text{ref}}} = \left[\frac{\int_0^\infty dk k^2 P_{\text{model}}(k, z=0) W_8^2(k)}{\int_0^\infty dk k^2 P_{\text{ref}}(k, z=0) W_8^2(k)} \right]^{1/2},$$

up to the common normalization factor. The corresponding S_8 ratio would also include $\sqrt{\Omega_{m,\text{model}}/\Omega_{m,\text{ref}}}$ if the matter-density normalization were varied. In this paper we quote only the amplitude proxy $\sqrt{P_{\text{geo}}}$. For the final branch, whose residual matter-power tilt is $\simeq 1.61\%$, the geometric-mean proxy is used only as a controlled amplitude-scale indicator and does not replace a likelihood analysis.

As a direct check on this geometric-mean proxy, we also evaluate the top-hat-windowed linear amplitude ratio from the archived dense $z=0$ matter-power arrays. The result is shown in Table VIII. This check tests whether $\sqrt{P_{\text{geo}}}$ is a faithful amplitude proxy for the final matter-power spectra. It is still not a weak-lensing, RSD, CMB, or combined likelihood analysis.

Branch	$\sqrt{P_{\text{geo}}}$	$\sigma_8/\sigma_{8,\text{ref}}$	Difference	Note
reference	1.000000	1.000000	$+0.000e+00$	reference
Newtonian-gauge benchmark	0.947239	0.949239	$+2.000e-03$	windowed check
drain-only	0.944047	0.944048	$+8.246e-07$	windowed check
B proxy	0.947239	0.949236	$+1.996e-03$	windowed check
D proxy	0.947239	0.949235	$+1.996e-03$	windowed check

TABLE VIII. Windowed σ_8 amplitude check from dense $z = 0$ matter-power arrays. The $\sqrt{P_{\text{geo}}}$ column uses the same $k = 0.01\text{--}0.30 h \text{ Mpc}^{-1}$ band used in the benchmark summaries, while $\sigma_8/\sigma_{8,\text{ref}}$ is computed with the top-hat window. This is an amplitude check only, not a likelihood analysis.

For the fiducial Newtonian-gauge branch,

$$P_{\text{geo}} \simeq 0.897145, \quad \sqrt{P_{\text{geo}}} \simeq 0.947177,$$

corresponding to an S_8 -facing proxy suppression of approximately

$$1 - \sqrt{P_{\text{geo}}} \simeq 5.28\%.$$

With the Planck 2018 reference amplitude $S_{8,\text{ref}} = 0.831$, this gives

$$S_{8,\text{proxy}} \simeq 0.787,$$

which is quoted only as a scale indicator, not as a data fit.

We also test two executable source-family proxy branches, denoted B and D, which represent alternative metric-source operator templates used only for numerical diagnostics. These branches remain in the same suppression regime. The source-B control proxy gives

$$P_{\text{geo}} \simeq 0.896795, \quad \sqrt{P_{\text{geo}}} \simeq 0.946993,$$

while the source-D candidate proxy gives

$$P_{\text{geo}} \simeq 0.896930, \quad \sqrt{P_{\text{geo}}} \simeq 0.947064.$$

Both correspond to an approximately 5.3% proxy amplitude suppression and satisfy the predefined S_8 -window gates. They do not validate a specific physical nonzero metric-source operator or physical nonzero metric forcing.

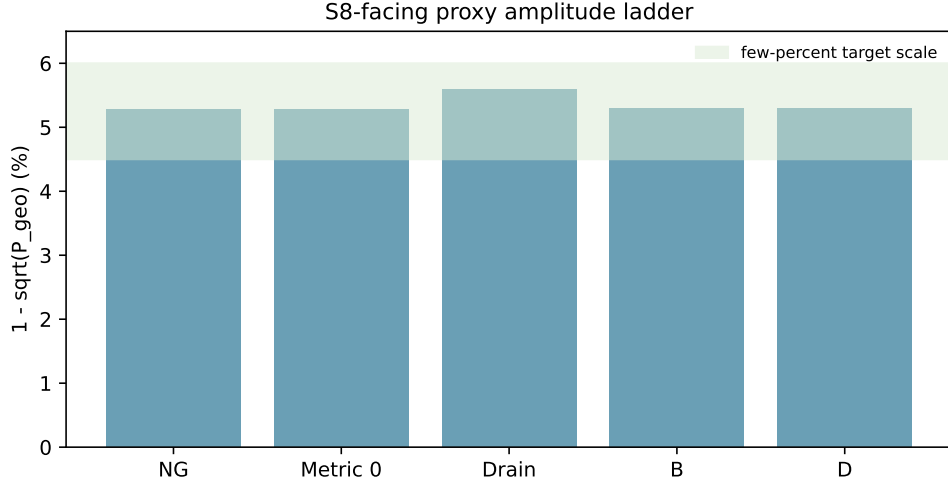


FIG. 6. S_8 -facing proxy amplitude ladder. The plotted quantity is $1 - \sqrt{P_{\text{geo}}}$ and is used only as a scale translation, not as a likelihood comparison. The Newtonian-gauge and executable source-family branches remain near a five-percent amplitude suppression scale.

Thus the Newtonian-gauge benchmark produces a controlled matter-power suppression in the amplitude range relevant to low-redshift S_8 comparisons. A statistical resolution of the S_8 tension would require a full likelihood analysis with weak-lensing, galaxy-clustering, CMB, BAO, and RSD data, including their nuisance parameters and covariance structure.

As a redshift-aware proxy check, the source-layer suppression is tracked across the tested redshift range and relaxes smoothly toward unity by $z \simeq 2$. The Newtonian-gauge CLASS branches currently enter the S_8 proxy table as $z = 0$ amplitude anchors. This is a ledger-level redshift-shape cross-check, not a redshift-dependent Newtonian-gauge CLASS validation or a full $f\sigma_8$ prediction.

J. Compressed observational amplitude comparison

As a minimal observable-facing check, we compare the amplitude scale of the CLASS benchmark with published compressed S_8 constraints. This is not a replacement for a DES, KiDS, HSC, RSD, Planck, or combined likelihood analysis. It only asks whether the matter-power suppression produced by the selection-fold drain lies in the amplitude range relevant to current low-redshift clustering measurements.

We use the Planck 2018 base- Λ CDM value as a reference amplitude,

$$S_{8,\text{ref}} = 0.831,$$

and define

$$S_8^{\text{comp}} = S_{8,\text{ref}} \sqrt{P_{\text{geo}}}.$$

For the fiducial Newtonian-gauge benchmark,

$$P_{\text{geo}} = 0.897145, \quad \sqrt{P_{\text{geo}}} = 0.947177,$$

which gives

$$S_8^{\text{comp}} \simeq 0.787.$$

This value lies in the same amplitude range as the low-redshift weak-lensing constraints from DES Y3, KiDS-1000, and HSC-Y3, while remaining below the Planck reference amplitude by construction. The windowed check in Table VIII shows that this geometric proxy agrees with the top-hat-windowed amplitude ratio at the quoted benchmark level. The corresponding windowed value differs only at the few- 10^{-3} level and does not change the compressed comparison.

The Planck point is used only to set the reference amplitude and is not included as an independent low-redshift likelihood point. We do not assign a compressed χ^2 to the comparison, because doing so would obscure the absence of tomographic kernels, nonlinear modeling, covariance matrices, and nuisance marginalization.

The compressed comparison shows that the benchmark suppression lies in the amplitude range relevant for low-redshift clustering comparisons. It does not establish a statistical resolution of the S_8 tension, because such a claim would require redshift-dependent Newtonian-gauge benchmark power spectra, weak-lensing kernels, nonlinear modeling, nuisance marginalization, covariance matrices, and a full likelihood pipeline.

VIII. LIMITATIONS AND INTERPRETATION

The present construction is an implementation benchmark, not a cosmological likelihood model. The source-level insertion tests show that the late-time selection-drain effect can be transmitted to the CLASS matter-power output with small residual scale dependence. They do not yet constitute a complete scalar-perturbation implementation.

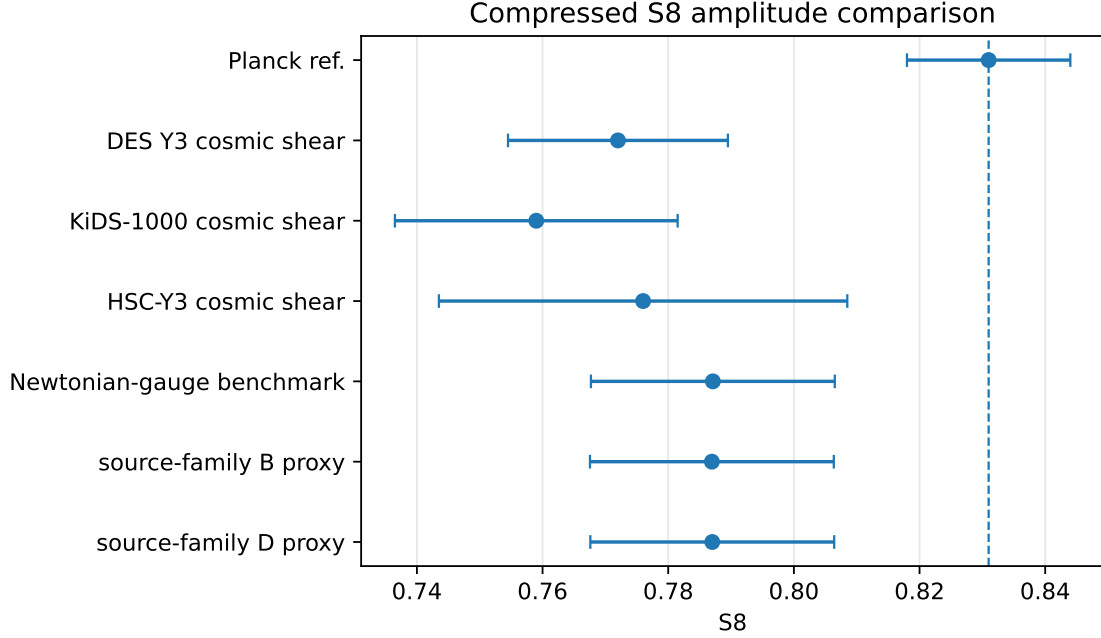


FIG. 7. Compressed S_8 amplitude comparison. The Planck point is used as the reference amplitude, while the DES Y3, KiDS-1000, and HSC-Y3 points are published low-redshift weak-lensing constraints. The benchmark branches are computed from $S_8^{\text{comp}} = S_{8,\text{ref}} \sqrt{P_{\text{geo}}}$. The displayed comparison is an amplitude-scale check only and is not a tomographic weak-lensing, RSD, CMB, or combined likelihood analysis.

The Newtonian-gauge branch registers $(\delta u, \delta x)$, verifies the matter-source insertion point through half-amplitude and drain-only controls, and produces a bounded final matter-power response. The metric-dependent terms are treated only as numerical coupling tests. In particular, the Λ_{met} scans establish finite behavior around the metric-on-zero branch, but

Compressed constraint	S_8^{obs}	S_8^{comp}	ΔS_8
DES Y3 cosmic shear, Λ CDM-optimized [2]	0.772 ± 0.018	0.787	+0.015
KiDS-1000 cosmic shear [3]	0.759 ± 0.022	0.787	+0.028
HSC-Y3 cosmic shear [5]	0.776 ± 0.033	0.787	+0.011

TABLE IX. Compressed S_8 amplitude comparison for the fiducial Newtonian-gauge CLASS branch. The model value is computed as $S_8^{\text{comp}} = S_{8,\text{ref}} \sqrt{P_{\text{geo}}}$ using $S_{8,\text{ref}} = 0.831$. This is an amplitude-scale comparison only, not a weak-lensing, RSD, CMB, or combined likelihood analysis.

do not validate a physical nonzero metric-forcing model or the full nonzero metric-source operator.

A further limitation is gauge completion. The dynamic response bridge is a calibrated first-order perturbation-response prototype; the coupled solver is a standalone growth-response replay; and the final in-code branch is a Newtonian-gauge benchmark. A gauge-complete Boltzmann treatment would require Newtonian-to-synchronous consistency, stress-energy closure, and a covariant perturbation implementation of the coupled u, x sector, comparable in spirit to complete modified-gravity or screened-scalar implementations [29–32].

The S_8 interpretation is likewise proxy-level. We translate the geometric mean matter-power suppression into an amplitude proxy through $S_{8,\text{model}}/S_{8,\text{ref}} \simeq \sqrt{P_{\text{geo}}}$ and record a redshift-aware ledger using the earlier source-layer redshift stability. We include only a compressed S_8 amplitude comparison against published low-redshift constraints. This comparison is useful for checking the scale of the effect, but it does not use the DES, KiDS, HSC, RSD, or Planck [33] likelihoods, and it does not include tomographic kernels, nonlinear corrections, baryonic feedback, intrinsic alignments, covariance matrices, or nuisance-parameter marginalization. It also does not constitute a full $f\sigma_8$ prediction or a fit to SH0ES [34]. The dense $z = 0$ spectra archived with this release support the direct top-hat-windowed amplitude check in Table VIII. A redshift-dependent Newtonian-gauge power-spectrum validation remains future work.

Finally, any likelihood-level test would require redshift-dependent Newtonian-gauge benchmark power spectra, weak-lensing or RSD data vectors, covariance matrices, nuisance-parameter marginalization, nonlinear scale treatment, and a consistent prior set. Such an analysis would also require an implementation suitable for standard sampling pipelines such as MontePython and Cobaya [35, 36]. These requirements define the likelihood pipeline needed for a future analysis. The present claim is therefore limited to a controlled Newtonian-gauge CLASS benchmark with an S_8 -facing proxy amplitude at the amplitude range relevant to low-redshift comparisons.

IX. CONCLUSION

We have developed a two-field selection-fold dark-drain mechanism for late-time matter-power suppression. The mechanism is driven by a fold transition in a selection branch

variable u , which entrains a physical scalar $x = \theta/M_{\text{Pl}}$ and modulates the cold-dark-matter mass through $m_c(x) = m_{c0} \exp[-\beta_x(x - x_{\text{ini}})]$. This produces a late-time drain of the cold-dark-matter density and motivates an S_8 -facing suppression of linear matter power.

The calculation proceeds from the background bridge to the source-layer insertion, response replacement, coupled growth replay, and Newtonian-gauge CLASS branch. The CLASS background bridge matches the reconstructed Friedmann branch with sub-percent expansion residuals and a derivative-level consistency check at the 7.68×10^{-10} level. The source-layer insertion verifies that the final CLASS mPk source insertion point is controllable, including a source-multiplier control with $P_{\text{on}}/P_{\text{off}} = 0.9604$ and negligible tilt. The response-derived quasistatic kernel, calibrated dynamic replacement, and standalone coupled growth-response replay preserve the stable suppression window and remain stable under local sensitivity and Picard-feedback tests.

The Newtonian-gauge CLASS implementation registers Newtonian response variables $(\delta u, \delta x)$ and transmits the dynamic source to final matter power. The fiducial Newtonian-gauge branch gives $P_{\text{geo}} \simeq 0.8971$, $P_{\text{min}} \simeq 0.8896$, $P_{\text{max}} \simeq 0.9116$, tilt $\simeq 1.61\%$, and $\mu_{\text{eff}} \simeq 0.9440\text{--}0.9454$. Metric-lambda hooks preserve the zero-lambda benchmark and support finite diagnostic numerical coupling, but no physical nonzero metric-forcing validation is claimed.

The resulting matter-power suppression corresponds to a roughly five-percent S_8 -facing amplitude shift under the proxy mapping $S_{8,\text{model}}/S_{8,\text{ref}} \simeq \sqrt{P_{\text{geo}}}$. This places the effect in the amplitude range relevant to low-redshift S_8 comparisons. The redshift-aware ledger further shows that the earlier source-layer suppression relaxes smoothly toward unity by $z \simeq 2$, while the Newtonian-gauge branches currently enter the S_8 proxy as $z = 0$ amplitude anchors.

The redshift relaxation behavior $P/P_{\text{off}} \rightarrow 1$ at $z \simeq 2$ has been verified for the source-layer bridge. The Newtonian-gauge CLASS benchmark currently provides the final in-code matter-power spectrum at $z = 0$; full redshift-dependent validation of that branch is left for future work.

A future extension could replace the phenomenological recordability terms in U_{sel} by microscopic influence-functional kernels [37, 38].

The work deliberately stops at a controlled Newtonian-gauge CLASS benchmark. It does not provide a gauge-complete Boltzmann closure, a physical validation of nonzero metric forcing, a specific physical nonzero metric-source operator, a full $f\sigma_8$ prediction, or a

likelihood-level data fit. Its main contribution is a reproducible bridge from a selection-fold dark-sector mechanism to a controlled CLASS matter-power suppression benchmark, with explicit scope boundaries and a likelihood pipeline roadmap.

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Appendix A: Reproducibility manifest

The curated public release accompanying this manuscript contains the L^AT_EX source, local table inputs, generated figures, dense $z = 0$ matter-power arrays used for the windowed σ_8 check, compressed S_8 comparison data, and scripts used to regenerate the plotted summaries. The release is organized as follows:

- `paper/`: manuscript source, bibliography, local tables, and figures.
- `data/background/`: background bridge summaries and diagnostic columns.
- `data/matter_power/`: matter-power ratios used in the benchmark figures.
- `data/matter_power_dense/`: dense $z = 0$ matter-power arrays for the reference and benchmark branches.
- `data/sigma8/`: windowed σ_8 amplitude-check table.
- `data/s8_compressed/`: compressed S_8 comparison table.
- `scripts/`: figure-generation, compressed-comparison, and windowed-amplitude scripts.
- `environment/`: package requirements used for plotting and table generation.

The package is intended to reproduce the benchmark tables and figures. It does not include a DES/KiDS/HSC/Planck likelihood pipeline, full Newtonian-gauge redshift-dependent

matter-power arrays, or a gauge-complete Boltzmann implementation.

- [1] T. M. C. Abbott *et al.* (DES Collaboration), Dark Energy Survey year 3 results: Cosmological constraints from galaxy clustering and weak lensing, *Phys. Rev. D* **105**, 023520 (2022).
- [2] L. F. Secco, S. Samuroff, E. Krause, *et al.* (DES Collaboration), Dark Energy Survey year 3 results: Cosmology from cosmic shear and robustness to modeling uncertainty, *Phys. Rev. D* **105**, 023515 (2022).
- [3] M. Asgari *et al.*, KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics, *Astron. Astrophys.* **645**, A104 (2021).
- [4] Y. Hikage *et al.*, Cosmology from cosmic shear power spectra with Subaru Hyper Suprime-Cam first-year data, *Publ. Astron. Soc. Jpn.* **71**, 43 (2019).
- [5] X. Li, T. Zhang, S. Sugiyama, *et al.*, Hyper Suprime-Cam year 3 results: Cosmology from cosmic shear two-point correlation functions, *Phys. Rev. D* **108**, 123518 (2023).
- [6] S. Alam *et al.* (eBOSS Collaboration), Completed SDSS-IV extended Baryon Oscillation Spectroscopic Survey: Cosmological implications from two decades of spectroscopic surveys, *Phys. Rev. D* **103**, 083533 (2021).
- [7] C. Heymans *et al.*, KiDS-1000 cosmology: Multi-probe weak gravitational lensing and spectroscopic galaxy clustering constraints, *Astron. Astrophys.* **646**, A140 (2021).
- [8] S. Joudaki *et al.*, KiDS+VIKING-450 and DES-Y1 combined: Cosmology with cosmic shear, *Astron. Astrophys.* **638**, L1 (2020).
- [9] S. Alam *et al.*, The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey, *Mon. Not. Roy. Astron. Soc.* **470**, 2617 (2017).
- [10] K. G. Wilson, Renormalization group and critical phenomena. i. renormalization group and the kadanoff scaling picture, *Phys. Rev. B* **4**, 3174 (1971).
- [11] S. Weinberg, *The Quantum Theory of Fields, Volume II: Modern Applications* (Cambridge University Press, Cambridge, 1996).
- [12] L. D. Landau and E. M. Lifshitz, *Statistical Physics* (Pergamon Press, Oxford, 1980).
- [13] V. I. Arnold, *Catastrophe Theory* (Springer, Berlin, 1992).
- [14] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, *Rev. Mod. Phys.* **75**, 715 (2003).

- [15] M. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition* (Springer, Berlin, 2007).
- [16] W. H. Zurek, Quantum Darwinism, *Nature Phys.* **5**, 181 (2009).
- [17] J. Rissanen, Modeling by shortest data description, *Automatica* **14**, 465 (1978).
- [18] P. D. Grunwald, *The Minimum Description Length Principle* (MIT Press, Cambridge, MA, 2007).
- [19] L. Amendola, Coupled quintessence, *Phys. Rev. D* **62**, 043511 (2000).
- [20] C. Wetterich, The cosmon model for an asymptotically vanishing time dependent cosmological constant, *Astron. Astrophys.* **301**, 321 (1995).
- [21] T. Koivisto, Growth of perturbations in dark matter coupled with quintessence, *Phys. Rev. D* **72**, 043516 (2005).
- [22] J. Valiviita, E. Majerotto, and R. Maartens, Instability in interacting dark energy and dark matter fluids, *JCAP* **2008** (07), 020.
- [23] M. B. Gavela, D. Hernandez, L. Lopez Honorez, O. Mena, and S. Rigolin, Dark coupling, *JCAP* **2009** (07), 034.
- [24] R. Bean, E. E. Flanagan, I. Laszlo, and M. Trodden, Constraining interactions in cosmology's dark sector, *Phys. Rev. D* **78**, 123514 (2008).
- [25] D. Blas, J. Lesgourgues, and T. Tram, The Cosmic Linear Anisotropy Solving System (CLASS). part II: Approximation schemes, *JCAP* **2011** (07), 034.
- [26] A. Lewis, A. Challinor, and A. Lasenby, Efficient computation of CMB anisotropies in closed FRW models, *Astrophys. J.* **538**, 473 (2000).
- [27] B. Hu, M. Raveri, N. Frusciante, and A. Silvestri, Effective field theory of cosmic acceleration: an implementation in CAMB, *Phys. Rev. D* **89**, 103530 (2014).
- [28] M. Zumalacarregui, E. Bellini, I. Sawicki, J. Lesgourgues, and P. G. Ferreira, hi_class: Horndeski in the Cosmic Linear Anisotropy Solving System, *JCAP* **2017** (08), 019.
- [29] J. Khoury and A. Weltman, Chameleon fields: Awaiting surprises for tests of gravity in space, *Phys. Rev. Lett.* **93**, 171104 (2004).
- [30] E. V. Linder, Cosmic growth history and expansion history, *Phys. Rev. D* **72**, 043529 (2005).
- [31] W. Hu and I. Sawicki, Models of $f(r)$ cosmic acceleration that evade solar-system tests, *Phys. Rev. D* **76**, 064004 (2007).
- [32] J. Noller and A. Nicola, Cosmological parameter constraints for modified gravity, *Phys. Rev.*

- D **104**, 123502 (2021).
- [33] N. Aghanim *et al.* (Planck Collaboration), Planck 2018 results. VI. cosmological parameters, *Astron. Astrophys.* **641**, A6 (2020).
 - [34] A. G. Riess *et al.*, A comprehensive measurement of the local value of the hubble constant, *Astrophys. J. Lett.* **934**, L7 (2022).
 - [35] B. Audren, J. Lesgourgues, K. Benabed, and S. Prunet, Conservative constraints on early cosmology with Monte Python, *JCAP* **2013** (02), 001.
 - [36] J. Torrado and A. Lewis, Cobaya: Code for bayesian analysis of hierarchical physical models, *JCAP* **2021** (05), 057.
 - [37] R. P. Feynman and F. L. Vernon, The theory of a general quantum system interacting with a linear dissipative system, *Ann. Phys.* **24**, 118 (1963).
 - [38] A. O. Caldeira and A. J. Leggett, Path integral approach to quantum brownian motion, *Physica A* **121**, 587 (1983).